

Generation Interconnection System Impact Study Report

For Queue Project Q16-0668

Nassau 138 kV

June 18, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0668. The Interconnection Customer proposes a 74.1 MW solar photovoltaic facility interconnecting at the Nassau 138 kV substation in the Virginia area. The request was received on August 23, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 343 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,051,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
NO TROY – OCONTO 138 kV	AVE 52 – 4 AC STA 138 kV	133.2	189	24.0
SPRINGBANK – OCONTO 138 kV	AVE 52 – 4 AC STA 138 kV L	128.3	206	24.0
SPORN – COULEE AVE 345 kV	INTERNATIONAL PAPER – HANGING ROCK 500 kV	125.2	1,109	8.4
BUTLER RIDGE – CINDER 138 kV	NEPTUNE – RGLAND 345 kV	119.1	222	8.2
HANGING ROCK SUBSTATION 500/345 kV Transformer	INTERNATIONAL PAPER – FLATLICK SUBSTATION 500 kV	114.6	787	7.4
BUTLER RIDGE – RAGLAND 138 kV	NEPTUNE – RAGLAND 345 kV	112.0	230	8.2
SHULLSURG – EDMONT 138 kV	HINTON AP – LOWMOOR 345 kV	111.5	234	18.0
S CHRISTIANBURG – HAPPY CANYON 138 kV	LOWMOOR – HINTON AP 345 kV Line	110.3	194	18.0
DAYTON 345/138 kV Autotransformer	CHATOM SUBSTATION – DAYTN 345 kV line	109.5	479	6.3
HAPPY CANYON – EDMONT 138 kV line	LOWMOOR – HINTON AP 345 kV	108.7	218	18.0
CHATOM 345/138kV Auto	DAYTON 345 kV/138 kV XFMR	107.0	494	6.3
MIDWAY – S CHRISTIANBURG 138 kV	HITON AP – LOWMOOR 345 kV	105.8	180	18.0
COULEE AVE – AMOS 345 kV	INTERNATIONAL PAPER – HANGING ROCK 500 kV	104.4	1,387	8.2
KIMPER – SPRIGG 138 kV	RAGLAND – NETUNE 345 kV	103.0	243	8.1
SPRIGG – CINDR SUBSTATION 138 kV line	RAGLAND – NEPTUNE 345 kV	102.6	237	8.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TREZEVANT – SHULLSBURG 138 kV	LOWMOOR – HINTON AP 345 kV line	102.5	307	24.9
DAYTON – HARRISONBURG 138 kV	DAYTON – CHATOM 345 kV line	101.6	487	6.3
MECKLENBURG POWER STATIN SUBSTATION – PERSON 345 kV	BEAR – CLOVER 345 kV	101.5	2,318	8.7
ROGUE – TREZEVANT 138 kV	HINTON AP – LOWMOOR 345 kV	101.3	333	24.9
ROGUE – LYNN 138 kV	HINTON AP SUBSTATION – LOWMOR 345 kV	101.2	337	24.9
FRSER – OU SPIRIT WIND FARM 138 kV L	HINTON AP – MANATEE 345 kV	101.0	227	14.1
LYNN – LOST NATION 138 kV	HINTON AP – LOWMOOR 345 kV line	100.9	342	24.9
LOST NATION – NO TROY 138 kV	HINTON AP – LOWMOOR 345 kV	100.8	347	24.9
CAIRO SUBSTATION – BECKER SUBSTATION 345 kV Line	DILLARD-ORANGE – BECK RD SUBSTATION 345 kV	100.5	1,193	10.4
DILLARD-ORANE – FISHKILL PLAINS 345 kV	DISMAL RIVER – BECKER 345 kV	100.4	2,830	7.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NO TROY – OCONTO 138 kV	Rebuild line with higher-capacity conductor	\$3,143,000	14% (\$433,000)
SPRINGBANK – OCONTO 138 kV line	Rebuild line with higher-capacity conductor	\$5,028,000	14% (\$693,000)
COULEE AVE – SPORN 345 kV line	Rebuild line with higher-capacity conductor	\$64,951,000	1% (\$516,000)
CINDER – BUTLER RIDGE 138 kV	Rebuild line with higher-capacity conductor	\$3,638,000	2% (\$72,000)
HANGING ROCK SUBSTATION 500/345 kV Transformer	Replace transformer with higher-rated unit	\$42,674,000	1% (\$407,000)
RAGLAND – BUTLER RIDGE 138 kV	Rebuild line with higher-capacity conductor	\$2,079,000	2% (\$41,000)
SHULLSBURG – EDGEMONT 138 kV	Rebuild line with higher-capacity conductor	\$3,661,000	13% (\$488,000)
S CHRISTIANSBURG – HAPPY CANYON 138 kV	Rebuild line with higher-capacity conductor	\$2,938,000	13% (\$392,000)
DAYTON 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$17,303,000	4% (\$782,000)
EDGEMONT – HAPPY CANYON 138 kV	Reconductor line	\$345,000	13% (\$46,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHATOM 345/138kV Auto	Replace transformer with higher-rated unit	\$8,943,000	4% (\$405,000)
MIDWAY – S CHRISTIANBURG 138 kV	Reconductor line	\$1,993,000	13% (\$266,000)
COULEE AVE – AMOS 345 kV	Reconductor line	\$23,767,000	1% (\$210,000)
KIMPER – SPRIGG 138 kV Line	Reconductor line	\$5,372,000	2% (\$100,000)
SPRIGG – CINDER 138 kV	Reconductor line	\$1,420,000	2% (\$26,000)
TREZEVANT – SHULLSBURG SUBSTATION 138 kV Line	Reconductor line	\$587,000	7% (\$43,000)
DAYTON – HARRISONBURG 138 kV	Reconductor line	\$553,000	4% (\$25,000)
MECKLENBURG POWER STATION – PERSON 345 kV	Reconductor line	\$48,771,000	1% (\$383,000)
ROGUE – TREZEVANT 138 kV	Reconductor line	\$4,852,000	7% (\$353,000)
LYNN – ROGUE 138 kV L	Reconductor line	\$4,938,000	7% (\$360,000)
FRSER – OU SPIRIT WIND FARM 138 kV L	Reconductor line	\$3,601,000	5% (\$196,000)
LYNN – LOST NATION 138 kV	Reconductor line	\$2,547,000	7% (\$186,000)
LOST NATION – NO TROY 138 kV	Reconductor line	\$1,256,000	7% (\$91,000)
CAIRO – BCKER 345 kV Line	Reconductor line	\$16,215,000	1% (\$167,000)
FISHKILL PLAINS – DILLARD-ORANGE 345 kV	Reconductor line	\$46,672,000	1% (\$370,000)
	Total allocated to project		\$7,051,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.